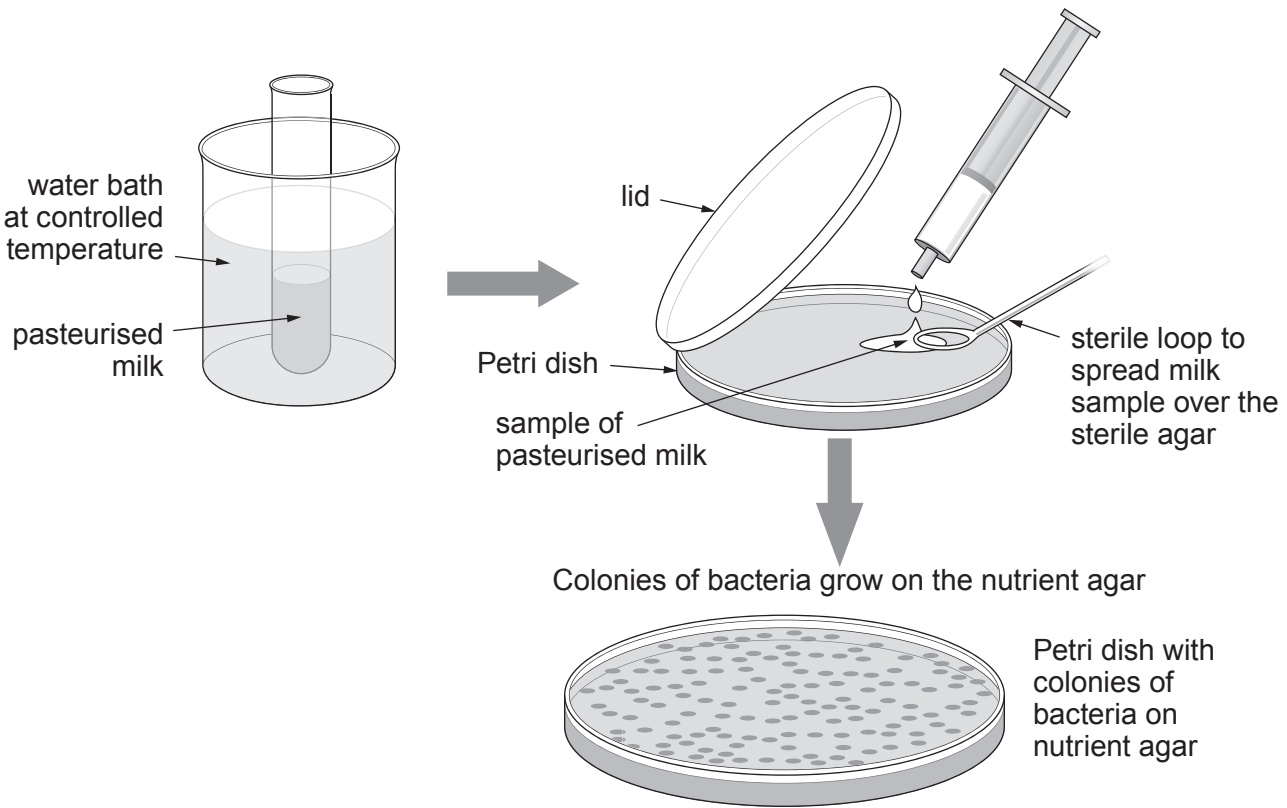


2. Fresh milk contains bacteria. Milk sold by supermarkets is pasteurised and has been heated to 72 °C for 20 seconds to kill most of the bacteria before it is bottled and stored.

Students investigated the growth of bacteria in pasteurised milk which they stored at temperatures of 10, 25 and 35 °C. They took samples of the milk at regular intervals and placed them on sterile nutrient agar to compare the growth of bacteria.

The diagram shows their procedure.



- (a) (i) State why it would be important for each milk sample to be of the same volume. [1]
-
- (ii) Describe how the loop used for spreading the sample could be sterilised and why this sterilisation was important. [2]
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-
-
- (iii) State what the students would do to the Petri dish to ensure aseptic conditions, after the milk sample had been spread on the agar. [1]
-



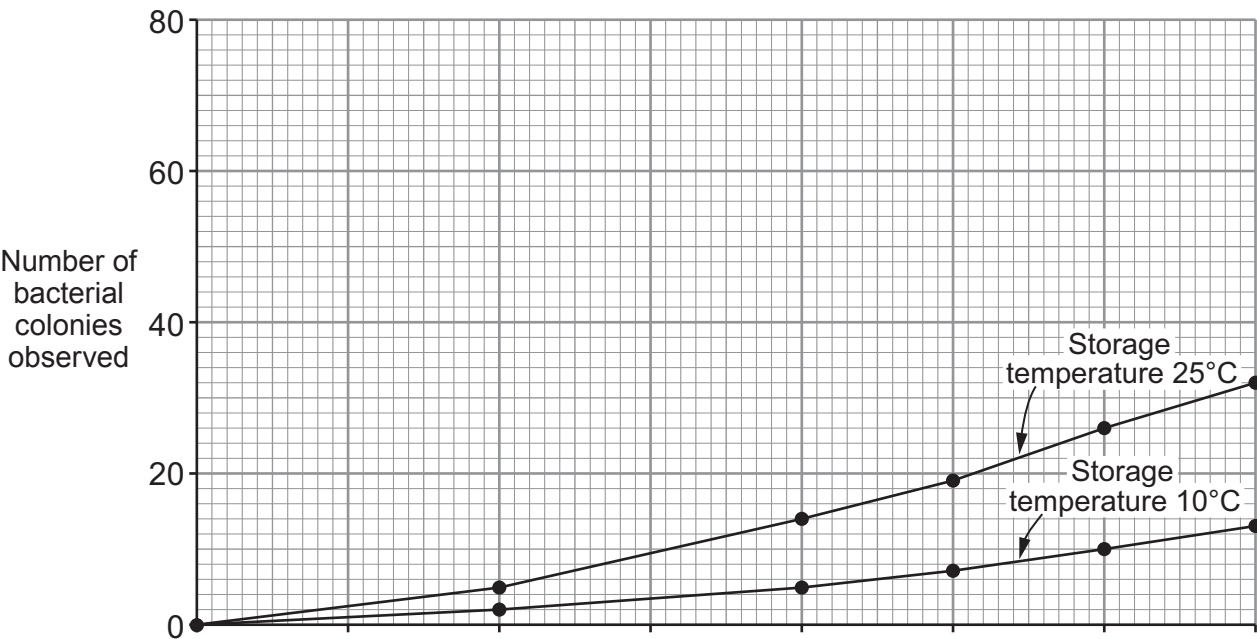
The results of the investigation are shown in the table and graph below.

Time of storage (hours)		Number of bacterial colonies observed		
		Storage temperature 10 °C	Storage temperature 25 °C	Storage temperature 35 °C
Start	0	0	0	0
	10	2	5	9
	20	5	14	28
	25	7	19	39
	30	10	26	50
	35	13	32	57

- (b) (i) Complete the graph by: [4]
- I. adding the scale and label for time;

II. plotting the results for 35 °C;

III. joining your plots with a ruler and labelling your line.



Examiner
only

(ii) Describe how the number of bacterial colonies changes over time and how this is affected by storage temperature. [2]

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(iii) I. Use the graph to estimate the difference in the number of colonies of bacteria between:
10 °C and 25 °C at 15 hours. [2]

Number of colonies

II. In the investigation, 50 colonies were found to have grown on the agar in a Petri dish. State how many bacteria would have been present in the sample of milk. Circle your answer. [1]

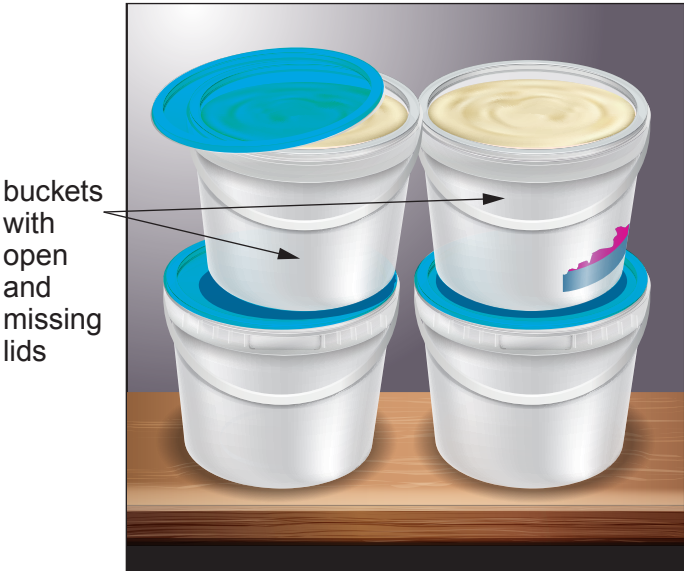
5 15 50 500

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(c) Food hygiene inspectors visited a restaurant in South Wales in the summer of 2010. They reported problems. The diagrams show some of their observations.

Buckets of yogurt made from pasteurised milk in a store cupboard at room temperature.



Close-up of a label on a bucket of yogurt.



From the problems they reported, the inspectors concluded that the level of hygiene at the restaurant was so low that it failed the inspection.

From the diagrams and your own knowledge explain the scientific reasons for the conclusion drawn by the inspectors. **Write your answers in the table below.** [3]

Problems reported by inspectors	Scientific reasons for inspectors' conclusion.
storage temperature	
lids of yogurt containers	
risk to customers' health	

